

THE ELSHOLTZ–TAO TYPE I SUM UNDER SELBERG’S EIGENVALUE CONJECTURE

ANONYMOUS

ABSTRACT. Assuming Selberg’s eigenvalue conjecture for $\Gamma_0(4dq^2)$ with even nebentypus modulo $2q$, uniformly for $d \geq 1$ and odd squarefree q coprime to d , we prove $\sum_{p \leq N} f_I(p) \ll N \log^2 N$, removing the $\log \log N$ from Elsholtz–Tao’s upper bound for Type I solutions of $4/p = 1/x + 1/y + 1/z$. The associated CM lifts of discriminant $-4d$ are uniformly separated in $\mathbb{H}$ ($\cosh \text{dist} \geq 3/2$). Sobolev duality and the Deshouillers–Iwaniec/Drappeau spectral large sieve give a per- d count complementary to a per- a Poisson–Weil count. Their savings degenerate linearly at $d = a$, without costing $\log \log N$. The proof is *conditional* on the stated spectral hypothesis and relative to cited inputs, including the inherited Elsholtz–Tao/internal-note reduction. It has received two rounds of internal review only, not external refereeing; that review did not re-check the inherited reduction. Exceptional-spectrum estimates leave a positive-width strip untreated, so this argument gives no unconditional improvement.

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1. INTRODUCTION

The Erdős–Straus equation is

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad x, y, z \in \mathbb{N}. \quad (1.1)$$

Following Elsholtz–Tao [ET], a solution is of Type I if $n \mid x$ and $(n, yz) = 1$, and of Type II if $n \mid y, z$ and $(n, x) = 1$. The counts retain coordinate order; for odd

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primes, $f(p) = 3f_I(p) + 3f_{II}(p)$ [ET, (1.4)]. Their Theorem 1.1 states

$$N \log^2 N \ll \sum_{p \leq N} f_I(p) \ll N \log^2 N \log \log N, \quad \sum_{p \leq N} f_{II}(p) \asymp N \log^2 N. \quad (1.2)$$

On p. 5 of arXiv:1107.1010v6, after Theorem 1.1, they attribute the double logarithm to Brun–Titchmarsh in very short progressions and conjecture its elimination. In §9, p. 36, a short modulus avoids the loss for Type II. Their parenthesis nevertheless says no similar trick seems available in the “Type II case” [sic]; reading this as Type I is the interpretation of [MN3, §1], not the literal quotation.

Put $\Delta = y^2(\partial_x^2 + \partial_y^2)$, so that $-\Delta$ is nonnegative. Here is the precise spectral hypothesis used in this note.

Hypothesis 1.1 (SEL). For every $d \geq 1$, every odd squarefree $q \geq 1$ with $(q, d) = 1$, and every even Dirichlet character χ modulo $2q$ (equivalently modulo q), $-\Delta$ on $L^2(\Gamma_0(4dq^2) \backslash \mathbb{H}, \chi)$ has no eigenvalue in $(0, 1/4)$.

The hypothesis covers the whole family; estimates are uniform in levels and characters. Constants are removed before applying the spectral estimate.

Theorem A. *Assume (SEL). Then*

$$\sum_{p \leq N} f_I(p) \ll N \log^2 N.$$

This is Theorem 8.2 below. It concerns an average upper bound, not existence of an Erdős–Straus representation for every prime.

The new input is uniform separation of the level- d CM points of discriminant $-4d$. Sobolev duality then pairs them with a mean-zero Poincaré series at a cost equal to the square root of their number times a second-order Sobolev norm, with no level loss. A spectral large sieve controls that norm under (SEL). The fixed- d count complements a fixed- a Weil count precisely at $d = a$; Lemma 8.1 sums the saving even though it degenerates linearly at that boundary. Only absolute errors averaged over d , not uniform per-packet relative errors, are used.

Unconditionally, the available estimates leave a positive-width strip next to $d = a$; a direct exponent comparison gives a scale about $7/32$ in $2\alpha - 1 + \gamma$, not a sharp boundary. This argument gives *no unconditional improvement* of (1.2); see §9.

Status and dependencies. Theorem A is CONDITIONAL on (SEL), with a proof relative to cited inputs. These are the ET reduction (Proposition 2.2, Lemma 2.8, and (8.1)–(8.2)); the internal note [MN3, Proposition 2.3 and proof of Theorem 3.8, step (1)]; Deshouillers–Iwaniec [DI, Theorem 2] and Drapeau [Dr, Proposition 4.7, arXiv numbering]; the spectral theorem; Weil’s bound; Brun–Titchmarsh; Selberg’s upper-bound sieve; and Pólya–Vinogradov. DI Theorem 5 enters the discussion of exceptional spectrum, not the conditional proof. The new argument received internal adversarial review (R111, rounds 1–2), not external refereeing. That review did not re-check the inherited ET/MN3 reduction; MN3 received a separate internal review. The book locators in [Iw] are inherited from the source note and have not been independently verified. Computational checks are not proofs.

Literature and scope. A limited literature search found no later removal of the Type I double logarithm; it was not an exhaustive priority search. Jia [Jia] is cited in ET’s discussion following Theorem 1.1, but for the Type II prime sum; the final arXiv version still calls the Type I double logarithm a technical limitation and conjectures its removal. Duke [Duke] and Liu–Masri–Young [LMY] address related

CM distributions, but do not supply the joint level-discriminant estimate needed here, with level comparable to the discriminant.

2. THE TYPE I SUM AND THE BAD REGION

Write $\mathcal{L} = \log N$ and work first with $N/2 < n \leq N$. The nonnegative weight

$$w_c(n) = \#\{(a, d, f) \in \mathbb{N}^3 : f \mid 4a^2d + 1, n = 4acd - f, n/4 < acd \leq 3n/4\} \quad (2.1)$$

satisfies $f_I(p) \leq 2 \sum_c w_c(p)$, by [ET, Proposition 2.2, Lemma 2.8]; the factor two restores the order of y, z . ET's map is $(x, y, z) = (abdn, acd, bcd)$, with $(abcd, n) = 1$ and $\gcd(a, b, c) = 1$. With $y \leq z$, it has

$$4abd = ne + 1, \quad ce = a + b, \quad 4acd = n + f, \quad ef = 4a^2d + 1, \quad bf = na + c, \quad (2.2)$$

and $a \leq b$, $an \leq bf \leq 5an/3$. Conversely, for every tuple in (2.1), put $e = (4a^2d + 1)/f$ and $b = ce - a$. Direct substitution gives all the identities (2.2), but only $b \geq a/2$, not necessarily $b \geq a$: indeed $0 < f \leq 2n$ and $bf = na + c$. We retain this larger set throughout. Writing $a = N^\alpha$, $b = N^\beta$, $c = N^\gamma$, we have

$$d \asymp N^{1-\alpha-\gamma}, \quad e \asymp N^{\beta-\gamma}, \quad f \asymp N^{1+\alpha-\beta}, \quad \beta \geq \alpha - O(\mathcal{L}^{-1}), \quad \alpha + \gamma \leq 1 + O(\mathcal{L}^{-1}). \quad (2.3)$$

All exponent regions below are understood up to these bounded dyadic errors.

Lemma 2.1 (SL₂ form; [MN3, Lemma 3.1]). *The identity $ef - 4a^2d = 1$ is equivalent to*

$$M = \begin{pmatrix} e & 2a \\ 2ad & f \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}), \quad n = 2cM_{21} - M_{22}.$$

If $T = \mathrm{diag}(1, d)$, then $\sigma_d(M) = TM^tT^{-1}$ is an anti-involution and $S_d = \{M \in \mathrm{SL}_2(\mathbb{Z}) : M_{21} = dM_{12}\}$ is its fixed set. It is stable under $M \mapsto \gamma M \sigma_d(\gamma)$ for $\gamma \in \Gamma_0(d)$. The positive Type I data also give the positive definite form $[f, 4ad, de]$ of discriminant $-4d$.

Proof. The determinant and linear-form assertions follow by multiplication. For $\gamma = \begin{pmatrix} p & t \\ r & s \end{pmatrix}$, $\sigma_d(\gamma) = \begin{pmatrix} p & r/d \\ dt & s \end{pmatrix}$ is integral exactly when $d \mid r$; reversing products proves stability. The form has discriminant $16a^2d^2 - 4def = -4d$ and positive leading coefficient. Conversely its coefficients recover f, a, e when the middle coefficient is divisible by $4d$. Positivity and this parity restriction are extra conditions, not invariants of all of S_d . $\square$

Remark 2.2 (Single-progression parametrisations). The generic elimination in [MN3, Lemma 3.2] produces the moduli

$$4ad, \quad 4bd, \quad 4ab, \quad 4acf, \quad 4cdf, \quad 4bcf', \quad 4cdf', \quad f' = 4bcd - n.$$

Its claim that these exhaust all three-coordinate single-progression parametrisations has status EVIDENCE: degenerate eliminants were not checked systematically. No completeness claim is used here. The following explicit fibres suffice. Fixing (a, c, f) forces $d \equiv -(4a^2)^{-1} \pmod{f}$ and puts n in one class modulo $4acf$. Fixing (c, d, f) puts a in $\rho_d(f) = \#\{x \pmod{f} : 4dx^2 + 1 \equiv 0 \pmod{f}\}$ classes, hence n in that many classes modulo $4cdf$. Fixing (a, b, c) fixes $e = (a + b)/c$; then $(e, 4ab) = 1$, $d \equiv (4ab)^{-1} \pmod{e}$, and $n = (4abd - 1)/e$ lies in one class modulo $4ab$. Every map from the free variable to n is injective. For fixed (a, b) there are at most $\tau(a + b)$ choices of c . The moduli $4ad, 4bd$ arise by varying c with the appropriate divisor fixed; swapping a, b gives the twins.

Proposition 2.3 (Bad-region geometry; [MN3, Proposition 3.3]). *In the ideal exponent domain $0 \leq \alpha \leq \beta$, $\alpha + \gamma \leq 1$, all seven displayed moduli have exponent at least $1 - \eta$ exactly on*

$$\mathcal{R}_{\text{bad}}(\eta) = \{\gamma \leq \eta, \alpha + \beta \geq 1 - \eta, \beta \leq 2\alpha + \gamma + \eta, \beta \leq 1 + \eta\}. \quad (2.4)$$

Its limiting $\gamma = \eta = 0$ slice has area $1/6$. There e, f are at least $\max(a, d)$ in exponent; for positive η this holds only up to $N^{-O(\eta)}$.

Proof. The seven exponents, in the order of the preceding list, are

$$1 - \gamma, \quad 1 - \gamma + \beta - \alpha, \quad \alpha + \beta, \quad 1 + 2\alpha + \gamma - \beta, \quad 2 - \beta, \quad 1 + 2\beta + \gamma - \alpha, \quad 2 - 2\alpha + \beta.$$

The second and last two inequalities are redundant, giving (2.4). At $\gamma = \eta = 0$ the slice is $\alpha + \beta \geq 1$, $\alpha \leq \beta \leq \min(2\alpha, 1)$, of area $\int_{1/3}^{1/2} (3\alpha - 1) d\alpha + \int_{1/2}^1 (1 - \alpha) d\alpha = 1/6$. Both β and $1 + \alpha - \beta$ are at least $\max(\alpha, 1 - \alpha)$ there. Perturbing the inequalities proves the last assertion. For our enlarged weights $b \geq a/2$ introduces only $O(\mathcal{L}^{-1})$ errors. $\square$

A positive area alone does not prove positive logarithmic mass or exclude other parametrisations. We use the region only to organise an upper bound.

2.1. Arithmetic estimates outside the bad region. We record the imported divisor estimate explicitly. Here $\omega(k)$ is the number of distinct prime divisors.

Proposition 2.4 (Coprimalty gain; [MN3, Proposition 2.3]). *Fix $l \geq 1$ and let $k \geq 1$, $U, V \geq 2$. (a) If $kV^2 \leq U^l$ and $U \geq \omega(k) + 2$, then*

$$\sum_{u \leq U, v \leq V} \tau(kuv^2 + 1) \ll_l \frac{\varphi(k)}{k} UV \log U.$$

(b) If $kU \leq V^l$ and $V \geq \omega(2k) + 2$, then the same sum is

$$\ll_l \frac{\varphi(k)}{k} UV \log V \left[1 + \min \left\{ \log(1 + k), \frac{k^{1/6} \log(2kU)}{\sqrt{U}} \right\} \right].$$

This is a proved input relative to [ET, Theorem 7.1, Corollary 7.4] and Pólya–Vinogradov. Its proof in [MN3, §2] keeps the indicator that divisors are coprime to k , rather than dropping it in the quadratic root bound. We do not re-prove that imported proposition.

Put $g(x) = x/\varphi(x)$ and write $x \asymp X$ for a dyadic interval $X < x \leq 2X$, allowing a separate initial interval containing 1.

Lemma 2.5 (Weighted divisor sums). *Uniformly in $A, B, D, F, X \geq 1$,*

$$(a) \quad \sum_{x \asymp X} g(x)^2 \ll X, \quad g(x) = \sum_{l|x} \frac{\mu^2(l)}{\varphi(l)}, \quad (2.5)$$

$$(b) \quad \sum_{\substack{a \asymp A, b \asymp B \\ a \leq 2b}} \tau(a + b) g(a) g(b) \ll AB \log(2B), \quad (2.6)$$

$$(c) \quad \sum_{d \asymp D} g(d) \sum_{f \asymp F} \frac{\rho_d(f)}{\varphi(f)} \ll D. \quad (2.7)$$

Proof. The identity in (2.5) follows on prime powers. Also $g^2 = 1 * h$ with $h \geq 0$, supported on squarefree integers, and $h(p) = (p/(p-1))^2 - 1 = O(1/p)$. Thus $\sum_l h(l)/l < \infty$, which proves the mean square and, by Cauchy-Schwarz, $\sum_{x \asymp X} g(x) \ll X$. The same divisor expansion gives $\sum_{x \leq X} 1/\varphi(x) \ll \log(2X)$.

For (2.6), fix a and retain $a \leq 2b$, so $a + b \leq 6B$. The elementary bound $\tau(n) \leq 2 \sum_{\delta|n, \delta \leq \sqrt{n}} 1$ gives, for squarefree l ,

$$\sum_{\substack{b \asymp B, l|b \\ a \leq 2b}} \tau(a+b) \leq 2 \sum_{\delta \leq \sqrt{6B}} \left(\frac{2B(l, \delta)}{l\delta} + 1 \right) \ll \frac{B}{l} \tau(l) \log(2B) + \sqrt{B}.$$

Multiply by $\mu^2(l)/\varphi(l)$ and sum $l \leq 2B$. The series $\sum_l \mu^2(l)\tau(l)/(l\varphi(l))$ converges and the remaining sum is $O(\sqrt{B} \log(2B))$. Summing with weight $g(a)$ proves (2.6).

For (2.7), set $\chi_d = (\frac{-4d}{\cdot})$ and $S_d(Y) = \sum_{r \leq Y} \chi_d(r)/r$, with $S_d(Y) = 0$ if $Y < 1$. This is a nonprincipal real Dirichlet character modulo $4d$. The prime-power root counts give

$$\rho_d(f) \leq (1 * \chi_d)(f), \quad \rho_d(kf) \leq 2^{\omega(k)} \rho_d(f) \quad (k \text{ squarefree}).$$

Indeed even moduli have no roots; for odd $p \nmid d$ the count at every positive power is $1 + \chi_d(p)$, and for $p \mid d$ it is zero. For $Y \geq 1$,

$$\sum_{Y < f \leq 2Y} (1 * \chi_d)(f) = Y S_d(2Y) + O(Y). \quad (2.8)$$

The floor-function error is $O(\sum_{r \leq 2Y} 1) = O(Y)$. We claim, uniformly in $Y \geq 1$,

$$\sum_{d \asymp D} |S_d(Y)|^2 \ll D. \quad (2.9)$$

For $Y \leq Y_0 = D/\log^2(2D)$ expand the square. Terms with rr' even vanish. If rr' is a square the inner d -sum is at most $O(D)$, and $\sum_{rr'=\square} (rr')^{-1} < \infty$ (write $r = su^2, r' = sv^2$ with s squarefree). Otherwise Pólya-Vinogradov for $d \mapsto (\frac{d}{rr'})$ bounds that sum by $O(\sqrt{rr'} \log(2rr'))$. The total is $O(D + Y \log(2Y)) = O(D)$. For $Y > Y_0$, partial summation and Pólya-Vinogradov modulo $4d$ give

$$S_d(Y) - S_d(Y_0) \ll \frac{\sqrt{D} \log(2D)}{Y_0} = O(1).$$

The last assertion is uniform for large D ; bounded D are handled directly by periodicity and partial summation. This proves (2.9); see [IK, Chapter 12] for the character-sum input. Finally the divisor expansion of $g(f)$, the root inequalities and (2.8) imply

$$\sum_{f \asymp F} \frac{\rho_d(f)}{\varphi(f)} \ll \sum_{k \geq 1} \frac{\mu^2(k) 2^{\omega(k)}}{k \varphi(k)} (|S_d(2F/k)| + 1).$$

When $F/k < 1$ the inner interval contains at most 1, so the same bound holds. Apply Cauchy-Schwarz using (2.5) and (2.9), and sum the convergent k -series. $\square$

Lemma 2.6 (Brun-Titchmarsh outside the bad region). *Fix $0 < \eta_1 < 1/2$. For $P \in \{ab, acf, cdf\}$, the number of tuples in (2.1), with arbitrary c , n prime in $(N/2, N]$, and $m_P \leq N^{1-\eta_1}$, is $O(\eta_1^{-1} N \mathcal{L}^2)$, where $m_{ab} = 4ab$, $m_{acf} = 4acf$, $m_{cdf} = 4cdf$.*

Proof. For $N \geq N_0(\eta_1)$, Brun–Titchmarsh [MV] on an interval of length $N/2$ gives $O(N/(\eta_1 \mathcal{L} \varphi(m)))$ primes per primitive class. A nonprimitive class contains at most one prime. By the injective fibres above, each fixed triple contributes at most

$$\rho_T \left(\frac{2N}{\eta_1 \mathcal{L} \varphi(m_T)} + 1 \right), \quad \rho_T = 1, 1, \rho_d(f),$$

respectively for ab, acf, cdf , with an inessential enlargement of the absolute constant. Use $\varphi(4x) \geq 2\varphi(x)$ and $\varphi(xy) \geq \varphi(x)\varphi(y)$. For acf the reciprocal-totient sum is $O(\mathcal{L}^3)$. For ab it is bounded by $\sum_{ab \leq N, a \leq 2b} \tau(a+b)/(\varphi(a)\varphi(b)) = O(\mathcal{L}^3)$, by (2.6) on $O(\mathcal{L}^2)$ boxes. For cdf , (2.7) gives $\sum_{df \leq N} \rho_d(f)/(\varphi(d)\varphi(f)) = O(\mathcal{L}^2)$; the c -sum supplies one more logarithm. The $+1$ errors total $O_\epsilon(N^{1-\eta_1+\epsilon})$, by the divisor bound and $\rho_d(f) \leq 2^{\omega(f)}$. Choose $\epsilon < \eta_1/2$. $\square$

Lemma 2.7 (Weighted Type I masses). *Let $\mathcal{L}^2 \leq A \leq N$ and $1 \leq D \leq N$. (a1) If $D \leq A$, then $\sum_{a \asymp A, d \asymp D} \tau(4a^2d+1)g(d) \ll AD\mathcal{L}$; if $D \geq A$, the same holds with $g(a)$ in place of $g(d)$. (a2) If also $A, D \geq N^{1/4}$, then*

$$\sum_{a \asymp A, d \asymp D} \tau(4a^2d+1)g(a)g(d) \ll AD\mathcal{L}. \quad (2.10)$$

(b) For every $A, D \geq 1$ and $F \leq 4A$,

$$\sum_{d \asymp D} g(d) \# \{(a, f) : a \asymp A, f \asymp F, f \mid 4a^2d+1\} \ll AD. \quad (2.11)$$

The last assertion also holds for e in place of f .

Proof. Expand $g(d) = \sum_{l \mid d} \mu^2(l)/\varphi(l)$ when $D \leq A$ and put $d = ld'$. Proposition 2.4, second assertion, with coefficient $4l$, linear range $2D/l$, quadratic range $2A$, and fixed exponent 40, gives

$$\sum_{a \leq 2A, d' \leq 2D/l} \tau(4la^2d'+1) \ll \frac{AD}{l} \mathcal{L}(1 + \log(1 + 4l)).$$

The required size inequalities are $8D \leq (2A)^{40}$ and $2A \geq \omega(8l) + 2$. If a range is smaller than 2, enlarge it to 2; then $8l \leq (2A)^{40}$ suffices and the loss is bounded since $l \leq 2D$. The resulting l -series with denominator $l\varphi(l)$ converges. For $D \geq A$, expand $g(a)$, put $a = ka'$, and use the first assertion of Proposition 2.4 with coefficient $4k^2$, ranges $2D, 2A/k$. Here $16A^2 \leq (2D)^{40}$ and $2D \geq \omega(4k^2) + 2$; after enlarging a small range use $16k^2 \leq (2D)^{40}$. The bound is $O((AD/k)\mathcal{L})$, summable against $1/\varphi(k)$.

For (2.10), expand both factors and write $a = ka'$, $d = ld'$, with coefficient $K = 4k^2l$. The part $k > A^{1/2}$ or $l > D^{1/2}$ is, for a sufficiently small fixed $\epsilon > 0$,

$$\ll N^\epsilon AD(A^{-1/2} + D^{-1/2}) \ll AD.$$

Here the divisor bound is uniform since $4a^2d+1 \ll N^3$, and $\sum_{k > T} 1/(k\varphi(k)) \ll T^{-1}$ follows from (2.5). In the remaining part, if $D \geq A$ use the first assertion of Proposition 2.4: its coefficient condition is $16lA^2 \leq (2D/l)^{40}$. If $D < A$ use the second assertion, whose condition is $8k^2D \leq (2A/k)^{40}$. These hold for all sufficiently large N , since the relevant large variable is at least $N^{1/8}$ and the left side is $O(N^3)$. The conditions involving $\omega(K)$ hold as well. Discarding $\varphi(K)/K \leq 1$, both bounds

are summable using

$$\sum_{k,l \geq 1} \frac{\mu^2(k)\mu^2(l)(1 + \log(1 + 4k^2l))}{kl\varphi(k)\varphi(l)} < \infty.$$

Finally a fixed (d, f) admits at most $\rho_d(f)(A/f + 1)$ values of a . For $f \leq 2F \leq 8A$ this is at most $9A\rho_d(f)/f$. Equation (2.7) proves (2.11). $\square$

2.2. What remains. Only the following unconditional step of the earlier LD reduction is needed.

Proposition 2.8 (Large- c reduction; [MN3, proof of Theorem 3.8, step (1)]). *For fixed small $\eta > 0$, the contribution of $c > N^\eta$ to $\sum_{N/2 < p \leq N} \sum_c w_c(p)$ is $O(\eta^{-1}N\mathcal{L}^2)$.*

Proof. Here $ad \ll N^{1-\eta}$. Brun-Titchmarsh in $n \equiv -f \pmod{4ad}$ and [ET, (8.1)–(8.2)] on product blocks give

$$\frac{N}{\eta\mathcal{L}} \sum_{ad \ll N} \frac{\tau(4a^2d + 1)}{\varphi(ad)} \ll \eta^{-1}N\mathcal{L}^2.$$

The classes are primitive since $f \mid 4a^2d + 1$ implies $(f, 4ad) = 1$, also for the enlarged weights. $\square$

It remains to count prime tuples with $c \leq N^\eta$ and

$$\min(4ab, 4acf, 4cdf) > N^{1-\eta_1} \quad (\eta_1 \geq \eta). \quad (2.12)$$

The moduli $4ad, 4bd$ are already $\gg N^{1-\eta}$ because $4acd \asymp N$ and $b \geq a/2$; the twin exponents are automatically large. Thus (2.12) lies in $\mathcal{R}_{\text{bad}}(\eta_1)$ up to $O(\mathcal{L}^{-1})$. Smooth upper-bound cells may be supported in $\mathcal{R}_{\text{bad}}(2\eta_1)$, where $e, f \geq N^{1/2-3\eta_1}$ up to absolute constants. This is the only remaining region; in particular no assertion of completeness from Remark 2.2 is needed.

3. $\text{SL}_2/\text{HEEGNER}$ REFORMULATION AND SEPARATION

For $d \geq 1$ define

$$\mathcal{F}_d = \{[A, B, C] \in \mathbb{Z}^3 : B^2 - 4AC = -4d, A > 0, 2d \mid B, d \mid C\}.$$

Its root map is $z_Q = (-B + i\sqrt{4d})/(2A)$. Every form here is positive definite. Type I forms form the subset

$$\mathcal{F}_d^I = \{[f, 4ad, de] : ef - 4a^2d = 1, e, f > 0, a \in \mathbb{Z}\}.$$

The unrestricted integer a makes the invariant set convenient; a positive box will restore $a > 0$. These are CM, or Heegner, forms with discriminant $-4d$ and level d , not forms satisfying the coprime Heegner hypothesis: $(4d, d)$ need not be 1. Type I forms are primitive, since $(f, 4ad) = 1$. The larger set $\mathcal{F}_d$ need not be primitive (e.g. $[2, 6, 6] \in \mathcal{F}_3$); Lemma 6.3 counts only the Type I subset.

Lemma 3.1 (Distance formula). *For positive definite forms Q, Q' of equal discriminant $D < 0$,*

$$\cosh \text{dist}_{\mathbb{H}}(z_Q, z_{Q'}) = 1 + \frac{\text{disc}(Q - Q')}{2|D|}.$$

Proof. Write $Q = [A, B, C]$, $Q' = [A', B', C']$. The formula $\cosh \text{dist}(z, w) - 1 = |z - w|^2 / (2\Im z \Im w)$ gives

$$\cosh \text{dist}(z_Q, z_{Q'}) - 1 = \frac{(BA' - B'A)^2 + |D|(A - A')^2}{2|D|AA'}.$$

Substituting $4AC = B^2 + |D|$ and $4A'C' = B'^2 + |D|$ into the discriminant of the difference gives

$$AA' \text{disc}(Q - Q') = (BA' - B'A)^2 + |D|(A - A')^2,$$

which proves the assertion. $\square$

Lemma 3.2 (Uniform separation). *Distinct $Q, Q' \in \mathcal{F}_d$ satisfy $\cosh \text{dist}_{\mathbb{H}}(z_Q, z_{Q'}) \geq 3/2$, independently of d .*

Proof. We have $4d^2 \mid (B - B')^2$ and $4d \mid 4(A - A')(C - C')$, hence $4d \mid \text{disc}(Q - Q')$. By Lemma 3.1, $\cosh \text{dist} - 1 \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. If it is zero, the roots coincide. A positive definite form of given negative discriminant is determined by its root: in $Q = A(X - zY)(X - \bar{z}Y)$ the discriminant fixes $A > 0$. Thus equality zero implies $Q = Q'$, which is excluded. $\square$

Separation persists on every subset, including parity and sieve subsets. It is a statement about all lifts in $\mathbb{H}$, not just a chosen list of representatives on a quotient. This distinction permits the uniform Sobolev argument of §4.

3.1. Coordinates, parity and the sieve group. Set

$$w_Q = z_Q/d = \frac{-2a + i/\sqrt{d}}{f} \quad (Q \in \mathcal{F}_d^I). \quad (3.1)$$

Dilation is a hyperbolic isometry. The set $\mathcal{F}_d$ is preserved by $Q \mapsto Q \circ \gamma$ for $\gamma \in \Gamma^0(d)$, where $\Gamma^0(d)$ means $d \mid \gamma_{12}$. Indeed

$$\begin{aligned} C' &= At^2 + Bts + Cs^2 \equiv 0 \pmod{d}, \\ B' &= 2Apt + B(ps + tr) + 2Crs \equiv 0 \pmod{2d}. \end{aligned}$$

for $\gamma = \begin{pmatrix} p & t \\ r & s \end{pmatrix}$. Roots transform by γ^{-1} . Conjugating by $z = dw$ identifies this with $\Gamma_0(d)$ on the w -side.

The smaller set $\mathcal{F}_d^I$ requires $4d \mid B$. On the z -side it is preserved by $\Gamma^0(2d) \cap \Gamma_0(2)$, since each term in the formula for B' is then divisible by $4d$. On the w -side this becomes $\Gamma_0(2d) \cap \Gamma^0(2)$. In particular, for squarefree odd $(q, d) = 1$, take

$$\Gamma'_q = \Gamma_0(2d) \cap \Gamma(2q), \quad u = w/(2q), \quad M = 4dq^2. \quad (3.2)$$

This preserves both the parity and the condition $q \mid n(Q) = cB - A$. To see the latter, the corresponding z -matrix is congruent to I modulo q , using $(q, d) = 1$; hence it preserves the coefficients of Q modulo q . It is important not to replace this group by $\Gamma_0(d) \cap \Gamma(2)$: for even d that group need not preserve $4d \mid B$ on the w -side.

In u -coordinates the effective group (adjoining $-I$, which acts trivially) is

$$\Gamma_{M,q} = \left\{ \pm \begin{pmatrix} p & t \\ r & s \end{pmatrix} \in \Gamma_0(M) : p \equiv s \equiv 1 \pmod{2q} \right\}.$$

Its cusp at infinity has width one, and it contains $\Gamma_1(M)$. For $q > 1$, its index in $\Gamma_0(M)$ is $\varphi(q)/2$; for $q = 1$ the group is simply $\Gamma_0(4d)$ and the index is 1. The quotient for $q > 1$ is $(\mathbb{Z}/q\mathbb{Z})^\times / \{\pm 1\}$. Consequently the spectral decomposition on

$\Gamma_{M,q}$ splits into the even-character spaces on $\Gamma_0(M)$ (with just the trivial space when $q = 1$). This explains precisely the family in (SEL).

3.2. Boxes and Poincaré series. For a smooth compactly supported weight in $a \asymp A, f \asymp F$, its expression on $\mathbb{H}$ is

$$\psi(w) = W_1\left(\frac{-\Re w}{2A\Im w\sqrt{d}}\right) W_2\left(\frac{1}{F\sqrt{d}\Im w}\right).$$

Its height is $\asymp (F\sqrt{d})^{-1}$, width $\asymp A/F$, and hyperbolic area $\asymp A\sqrt{d}$. For spectral estimates use a partition in $(\log f, \log(a/f))$. In the u -coordinate the cells then have the product shape

$$\psi(u) = \phi(x/\lambda)W(y/Y), \quad \lambda \asymp \frac{A}{qF}, \quad Y \asymp \frac{1}{qF\sqrt{d}}, \quad (3.3)$$

where $\text{supp } \phi \subset [-2, -1]$ and $\text{supp } W \subset [1, 2]$. Fixed smooth enlargements cover the original boxes by positivity. In particular $Y/\lambda \asymp (A\sqrt{d})^{-1}$. In the bad region this is a negative power of N , so the condition $Y \leq \lambda L_*^{-3}$ of Proposition 5.1 holds, since its local logarithm $L_* \ll \mathcal{L}$.

Here is the counting identity with the stabilisers made explicit. Let $\tilde{\Lambda}$ be the set of points arising from an invariant subset of $\mathcal{F}_d$ in the chosen coordinate, let Γ be its group, and let its cusp width at infinity be h . Set

$$\psi_{\text{per}}(z) = \sum_{m \in \mathbb{Z}} \psi(z + mh), \quad P_\psi(z) = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} \psi_{\text{per}}(\gamma z).$$

Write $\Lambda = \Gamma \backslash \tilde{\Lambda}$ and $e_z = |\Gamma_z / (\Gamma \cap \{\pm I\})|$. This set is finite: there are finitely many integral-form classes of fixed discriminant, and Γ has finite index in the appropriate conjugate of $\text{SL}_2(\mathbb{Z})$. With $d\mu = dx dy/y^2$ and $\#\Lambda = \sum_{z \in \Lambda} e_z^{-1}$, unfolding gives

$$N_d(\psi) := \sum_{w \in \tilde{\Lambda}} \psi(w) = \sum_{w \in \Gamma_\infty \backslash \tilde{\Lambda}} \psi_{\text{per}}(w) = \sum_{z \in \Lambda} \frac{P_\psi(z)}{e_z}. \quad (3.4)$$

Each lift occurs in exactly e_z cosets, cancelled by its weight; root-map injectivity ensures that forms are counted once. Also

$$\langle P_\psi \rangle = \frac{1}{\text{vol}(\Gamma \backslash \mathbb{H})} \int_{\mathbb{H}} \psi d\mu.$$

Periodisation is essential: there is no restriction $\lambda < 1$. The zero Fourier mode when $\lambda > 1$ will contribute a genuine term λ^2/Y to the variance, not an ignorable boundary error.

Finally the Fricke isometry exchanges the two divisor coordinates:

$$-\frac{1}{dw_Q} = \frac{2a + i/\sqrt{d}}{e}.$$

After reflection in the imaginary axis this is the point for $[e, 4ad, df]$. Thus one can use the smaller divisor in a suitable cusp. When e, f are exchanged the original prime functional becomes $cB - C/d$, rather than $cB - A$; the sieve density must be checked for that functional too, as in §6.

4. SOBOLEV DUALITY

Throughout this section Γ denotes the *effective* image in $\mathrm{PSL}_2(\mathbb{R})$ of a finite-index subgroup of $\mathrm{SL}_2(\mathbb{Z})$, or a conjugate thereof. In particular, kernel sums never count both signs of a matrix. Write $V = \mathrm{vol}(\Gamma \backslash \mathbb{H})$. For a Γ -invariant discrete set $\tilde{\Lambda}$ with finitely many orbits, put

$$e_z = |\Gamma_z|, \quad \nu_\Lambda = \sum_{z \in \Gamma \backslash \tilde{\Lambda}} e_z^{-1} \delta_z, \quad \#\Lambda = \sum_{z \in \Gamma \backslash \tilde{\Lambda}} e_z^{-1}.$$

These weights are necessary at elliptic points.

Lemma 4.1 (Resolvent kernel; standard). *There is a nonnegative continuous radial function k_0 on $\mathbb{H}$, with*

$$k_0(r) \ll (1+r)e^{-s_0 r}, \quad s_0 = \frac{1+\sqrt{5}}{2},$$

such that the kernel of $(1-\Delta)^{-2}$ on $L^2(\Gamma \backslash \mathbb{H})$ is $K(z, w) = \sum_{\gamma \in \Gamma} k_0(\mathrm{dist}(z, \gamma w))$. The sum converges absolutely.

Proof. The Selberg–Harish-Chandra multiplier is $h(t) = (5/4 + t^2)^{-2}$; see [Iw, §1.8] for the transform and its inversion, and [Iw, Theorem 1.14] for automorphisation. It is even, holomorphic in $|\Im t| < \sqrt{5}/2$, and integrable against $t \tanh(\pi t) dt$. Inversion therefore gives a continuous bounded point-pair kernel, with $k_0(0) = (4\pi)^{-1} \int_{\mathbb{R}} h(t) t \tanh(\pi t) dt$. For completeness, the decay can also be read directly from the resolvent kernel. For $s > 1$, the kernel of $(-\Delta + s(s-1))^{-1}$ on $\mathbb{H}$ is

$$R_s(r) = \frac{1}{2\pi\sqrt{2}} \int_r^\infty \frac{e^{-(s-1/2)v}}{\sqrt{\cosh v - \cosh r}} dv.$$

Taking the negative derivative in $s(s-1)$, at $s = s_0$, gives

$$k_0(r) = \frac{1}{2\pi\sqrt{2}(2s_0-1)} \int_r^\infty \frac{v e^{-(s_0-1/2)v}}{\sqrt{\cosh v - \cosh r}} dv.$$

This proves positivity. For $r \geq 1$, substitute $v = r + u$ and use $\cosh(r+u) - \cosh r \gg e^r(e^u - 1)$; the resulting integral is $\ll (1+r)e^{-s_0 r}$. At $r = 0$ the numerator cancels the singularity, giving the same boundedness and continuity as inversion. An orbit of a discrete cofinite group has $O_{z,w,\Gamma}(e^R)$ points, with multiplicity, within distance R . Since $s_0 > 1$, the periodisation is absolutely convergent. The automorphic kernel theorem identifies it with the multiplier h , which is precisely $(1-\Delta)^{-2}$. $\square$

Lemma 4.2 (Separated sets). **PROVED.** *If distinct points of $\tilde{\Lambda}$ have distance at least $r_0 > 0$, then*

$$\|\nu_\Lambda\|_{H^{-2}}^2 := \langle \nu_\Lambda, (1-\Delta)^{-2} \nu_\Lambda \rangle \leq C(r_0) \#\Lambda,$$

where $C(r_0)$ is independent of the group and the set.

Proof. Using Lemma 4.1 and cancelling the stabiliser multiplicity in one variable gives

$$\langle \nu_\Lambda, (1-\Delta)^{-2} \nu_\Lambda \rangle = \sum_{z \in \Gamma \backslash \tilde{\Lambda}} \frac{1}{e_z} \sum_{w \in \tilde{\Lambda}} k_0(\mathrm{dist}(z, w)).$$

Indeed each lift of w occurs e_w times in the kernel sum. The balls of radius $r_0/2$ around distinct lifts are disjoint. Those with centres within distance R of z lie in the

ball of radius $R + r_0/2$; comparison with its area $4\pi \sinh^2((R + r_0/2)/2)$ bounds their number by $O_{r_0}(e^R)$. Thus each inner sum is at most $O_{r_0}(\sum_{m \geq 0} (1+m)e^{(1-s_0)m}) = O_{r_0}(1)$. All rearrangements are justified by positivity and this convergent bound. $\square$

Proposition 4.3 (Uniform Sobolev duality). *PROVED. Suppose P is smooth and $P, \Delta P, \Delta^2 P \in L^2(\Gamma \backslash \mathbb{H})$. Put $\langle P \rangle = V^{-1} \int P d\mu$ and $P_0 = P - \langle P \rangle$. Under Lemma 4.2's separation hypothesis,*

$$\left| \sum_{z \in \Gamma \backslash \tilde{\Lambda}} \frac{P(z)}{e_z} - \# \Lambda \langle P \rangle \right| \leq C(r_0)^{1/2} (\# \Lambda)^{1/2} \|(1 - \Delta)P_0\|_2.$$

Proof. Lemma 4.2 says that $(1 - \Delta)^{-1} \nu_\Lambda \in L^2$ with squared norm at most $C(r_0) \# \Lambda$. Self-adjointness, first for smooth regularisations of the point masses and then by Sobolev duality, gives

$$\nu_\Lambda(P_0) = \langle (1 - \Delta)P_0, (1 - \Delta)^{-1} \nu_\Lambda \rangle.$$

Point evaluation agrees with this distributional pairing by local Sobolev embedding in dimension two. Cauchy–Schwarz proves the claim. $\square$

Corollary 4.4 (Heegner forms). *PROVED. Let $\mathcal{F} \subset \mathcal{F}_d$ be invariant under a subgroup $\Gamma \subset \Gamma_0(d)$ acting on $w_Q = z_Q/d$, with finitely many orbits. Proposition 4.3 holds with an absolute constant for its root set. For $\psi \in C_c^\infty(\mathbb{H})$, the periodised Poincaré series of §3 satisfies its integrability hypotheses, the counting identity (3.4), and $\langle P_\psi \rangle = V^{-1} \int_{\mathbb{H}} \psi d\mu$. The same holds after conjugating the group by a real dilation.*

Proof. The dilation is a hyperbolic isometry, so Lemma 3.2 supplies the absolute separation $r_0 = \operatorname{arccosh}(3/2)$. Unfold the group sum, cancelling elliptic stabilisers as in Lemma 4.2, and then unfold translations. The root map is injective at fixed discriminant, so this counts forms, not just roots. Unfolding the integral proves the mean formula. The Poincaré series and its Laplacian iterates are smooth and vanish sufficiently high in every cusp: for a nonstabilising coset the transformed height tends to zero, below the support of ψ , whereas at infinity the stabilising coset eventually lies above its support. Thus they satisfy the integrability assumptions of Proposition 4.3. $\square$

5. THE SPECTRAL LARGE SIEVE AND THE VARIANCE OF BOX POINCARÉ SERIES

Use the effective group $\Gamma_{M,q}$ of §3, $M = 4dq^2$, with squarefree $(q, 2d) = 1$ and cusp width one. Write $h_q = [\Gamma_0(M) : \Gamma_{M,q}]$: $h_1 = 1$ and $h_q = \varphi(q)/2$ for $q > 1$. Its h_q character spaces are the even nebentypus spaces modulo q .

The imported spectral estimates. We specify normalisations to distinguish a cusp width from the parameter in the large sieve. At a singular cusp $\mathfrak{a}$, choose a scaling matrix giving period one, and write the expansion of an orthonormal weight-zero Maass form as

$$u_j(\sigma_{\mathfrak{a}} z) = \sqrt{y} \sum_{n \neq 0} \rho_{j\mathfrak{a}}(n) K_{it_j}(2\pi|n|y) e^{2\pi i n x}. \quad (5.1)$$

This is DI's normalisation [DI, (1.34), p. 231]. The Eisenstein coefficients $\rho_{\text{ca}}(n, t)$ will use the identical $\sqrt{y}K_{it}$ convention. The DI cusp parameter for $\Gamma_0(M)$ is

$$\mu(\mathfrak{a}) = \frac{(v, M/v)}{M} \quad \text{if } \mathfrak{a} \sim u/v, \ v \mid M, \ (u, v) = 1; \quad \mu(\infty) = M^{-1}.$$

It is *not* the literal translation width at infinity, which is one. For $K \geq 1$, $H \geq 1/2$, and a_n supported on $H < n \leq 2H$, DI [DI, Theorem 2, (1.29)–(1.30), p. 230] bounds the cuspidal expression

$$\sum_{|t_j| \leq K} \frac{1}{\cosh(\pi t_j)} \left| \sum_n a_n \rho_{j\mathfrak{a}}(n) \right|^2$$

and its continuous-spectrum analogue by $(K^2 + \mu(\mathfrak{a})H^{1+\varepsilon}) \sum_n |a_n|^2$, for trivial nebentypus. With an even character of conductor $q_0 \mid q$, the bound is

$$\ll_{\varepsilon} (K^2 + q_0^{1/2} \mu(\mathfrak{a})H^{1+\varepsilon}) \sum_n |a_n|^2. \quad (5.2)$$

Here the continuous analogue means $\sum_c (4\pi)^{-1} \int_{-K}^K \cosh(\pi t)^{-1} |\sum_n a_n \rho_{\text{ca}}(n, t)|^2 dt$, summed over all singular cusps. This is [Dr, Proposition 4.7, (4.24)–(4.25)], in the arXiv numbering. Drappeau uses Whittaker coefficients: since $W_{0,it}(4\pi|n|y) = 2\sqrt{|n|y}K_{it}(2\pi|n|y)$, $\rho^{\text{DI}}(n) = 2\sqrt{|n|}\rho^{\text{Dr}}(n)$. Thus his $\sqrt{n}\rho^{\text{Dr}}(\pm n)$ is exactly our coefficient up to the constant 2; his factor $(1 + |t|)^{\pm\kappa}$ is one because $\kappa = 0$. These estimates hold separately for either sign of n .

For later comparison, DI [DI, Theorem 5, (1.38), p. 232] states, for trivial character, $X \geq 1$, and exceptional parameters $t_j = i\sigma_j$, $0 < \sigma_j < 1/2$,

$$\sum_{j \text{ exc}} X^{2\sigma_j} \left| \sum_{H < n \leq 2H} a_n \rho_{j\mathfrak{a}}(n) \right|^2 \ll_{\varepsilon} (1 + \sqrt{\mu(\mathfrak{a})HX})(1 + \sqrt{\mu(\mathfrak{a})H^{1+\varepsilon}}) \|a\|_2^2. \quad (5.3)$$

The nebentypus analogue is [Dr, Lemma 4.8], with second factor $1 + (q_0\mu(\mathfrak{a})H)^{1/2+\varepsilon}$. The displayed DI statements have been checked against the scan; the Drappeau numbering and coefficient conversion refer to arXiv:1504.05549, not to an unchecked published numbering.

The box variance. Take $\phi \in C_c^\infty([-2, -1])$, $W \in C_c^\infty([1, 2])$, and

$$\psi(u) = \phi(x/\lambda)W(y/Y), \quad L_* = \log(2 + (\lambda Y)^{-1} + Y^{-1} + M).$$

Assume $\lambda > 0$ and $Y \leq \lambda L_*^{-3}$. Periodisation is always understood; no restriction $\lambda < 1$ is imposed. For nonnegative nonzero weights its area is $\asymp \lambda/Y$. In applications $\lambda \asymp A/(qF)$, $Y \asymp 1/(qF\sqrt{d})$, and $L_* \ll \mathcal{L}$.

Proposition 5.1 (Box variance). CONDITIONAL on (SEL). On $\Gamma_{M,q} \backslash \mathbb{H}$,

$$\|(1 - \Delta)(P_\psi - \langle P_\psi \rangle)\|_2^2 \ll_{\phi, W, \varepsilon} L_*^C \left\{ \frac{\lambda}{Y} \left(1 + \frac{q^{1/2}}{M} (L_*/\lambda)^{1+\varepsilon} \right) + \frac{\lambda^2}{Y} \right\}. \quad (5.4)$$

The constant is uniform in d, q, λ, Y , and depends on only finitely many seminorms of the two fixed weights.

Proof. Reduction and spectral expansion. Differentiation commutes with periodisation and

$$(1 - \Delta)\psi = \phi(x/\lambda)(W(v) - v^2 W''(v)) - (Y/\lambda)^2 \phi''(x/\lambda) v^2 W(v), \quad v = y/Y.$$

Moreover $\int \Delta\psi d\mu = 0$. Since $Y/\lambda \leq 1$, it suffices to bound $\|P_\psi - \langle P_\psi \rangle\|_2^2$ by the right side of (5.4), for arbitrary fixed smooth weights of this type. For each even $\chi \bmod q$, let

$$P_\chi = \sum_{\gamma \in \Gamma_\infty \backslash \Gamma_0(M)} \overline{\chi(\gamma)} \psi_{\text{per}}(\gamma u).$$

Character orthogonality gives $P_\psi = h_q^{-1} \sum_\chi P_\chi$ and

$$\|P_\psi - \langle P_\psi \rangle\|_{\Gamma_{M,q}}^2 = \frac{1}{h_q} \sum_\chi \|P_\chi - \mathbf{1}_{\chi=1} \langle P_1 \rangle\|_{\Gamma_0(M), \chi}^2. \quad (5.5)$$

The spectral expansion, with Eisenstein measure $dt/(4\pi)$, is the usual Parseval formula [Iw, Theorem 7.3] (with the character spaces and singular cusps as in [Dr, §4.1]). *This is the sole use of (SEL):* after removing the constant, all discrete parameters are real, and there is no nonconstant residual spectrum. Thus the squared norm for a fixed character is the sum of $|\langle P_\chi, u_j \rangle|^2$ over cusp forms and $(4\pi)^{-1} \sum_c \int_{\mathbb{R}} |\langle P_\chi, E_c(\cdot, 1/2 + it) \rangle|^2 dt$.

Unfolding and Mellin inversion. Write $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(v) e^{-2\pi i \xi v} dv$. The periodised box has Fourier coefficient $\lambda \widehat{\phi}(\lambda n) W(y/Y)$, even when $\lambda > 1$. Using (5.1), unfolding gives

$$\langle P_\chi, u_j \rangle = \sum_{n \neq 0} \overline{\rho_j(n)} \lambda \widehat{\phi}(\lambda n) I_{t_j}(n), \quad I_t(n) = \int_0^\infty W(y/Y) y^{-3/2} K_{it}(2\pi|n|y) dy.$$

Set

$$G_t(s) = \Gamma((s+it)/2) \Gamma((s-it)/2), \quad \mathcal{W}(s) = \int_0^\infty W(v) v^{-3/2-s} dv.$$

Write also

$$B_j(s) = \sum_{n \neq 0} \overline{\rho_j(n)} \lambda \widehat{\phi}(\lambda n) |n|^{-s}.$$

The function $\mathcal{W}$ is entire and rapidly decreasing vertically, uniformly on fixed strips. Mellin inversion of the Bessel function gives, for $\sigma > 0$,

$$\langle P_\chi, u_j \rangle = \frac{Y^{-1/2}}{8\pi i} \int_{(\sigma)} G_{t_j}(s) \mathcal{W}(s) (\pi Y)^{-s} B_j(s) ds. \quad (5.6)$$

Absolute convergence follows from the rapid decay of $\widehat{\phi}$, the polynomial growth of Fourier coefficients, and Stirling's formula.

Uniform spectral bounds. Put $\sigma_0 = 1/L_*$. Stirling's formula, including the simple poles of Γ , gives

$$\begin{aligned} |G_t(\sigma_0 + iv)| &\ll L_*^2 e^{-\pi|v|/2} & (|t| \leq 1), \\ |G_t(-1 + iv)| &\ll e^{-\pi|t|/2} (1 + |t|)^{-2} (1 + |v|)^4 & (|t| \geq 1). \end{aligned}$$

For the second bound, if $|v| \leq |t|/2$ both gamma arguments have imaginary part comparable to $|t|$; otherwise $1 + |t| \ll 1 + |v|$ and the exponential is $e^{-\pi \max(|v|, |t|)/2}$. On this line both real parts are $-1/2$, a fixed distance from gamma poles. Since $Y^{-\sigma_0} \ll 1$, Cauchy-Schwarz in (5.6) yields, for $|t_j| \leq 1$,

$$|\langle P_\chi, u_j \rangle|^2 \ll Y^{-1} L_*^4 \int_{\mathbb{R}} |\mathcal{W}(\sigma_0 + iv)| |B_j(\sigma_0 + iv)|^2 dv.$$

For $|t_j| \geq 1$, shift to $\Re s = -1$, crossing only $s = \pm it_j$. The residue contribution is

$$\frac{Y^{-1/2}}{2} \sum_{\pm} \Gamma(\pm it_j) \mathcal{W}(\pm it_j) (\pi Y)^{\mp it_j} B_j(\pm it_j).$$

Let $\tilde{B}_j(s) = (\pi Y)^{-s} B_j(s)$. Using $|\Gamma(it)|^2 = \pi/(|t| \sinh(\pi|t|))$, for any fixed large B the squared coefficient is at most

$$\begin{aligned} & \frac{C_B Y^{-1}}{\cosh(\pi t_j)} \left\{ (1 + |t_j|)^{-B} \sum_{\pm} |B_j(\pm it_j)|^2 \right. \\ & \quad \left. + (1 + |t_j|)^{-4} \int_{\mathbb{R}} |\mathcal{W}(-1 + iv)| (1 + |v|)^4 |\tilde{B}_j(-1 + iv)|^2 dv \right\}. \end{aligned} \quad (5.7)$$

The large sieve, including both tails. Split n by sign and into dyadic blocks $H < |n| \leq 2H$, $H = 2^{i-1}$, $i \geq 0$. With $w_H = 1 + |\log(\lambda H)|$, one has $\sum_H w_H^{-2} \ll 1$. Cauchy-Schwarz across blocks and (5.2) imply

$$\mathcal{S}(K; b) := \sum_{|t_j| \leq K} \frac{|\sum_{n \neq 0} \overline{\rho_j(n)} b_n|^2}{\cosh(\pi t_j)} \ll_{\varepsilon} \sum_H w_H^2 (K^2 + q^{1/2} M^{-1} H^{1+\varepsilon}) \|b^{(H)}\|_2^2.$$

Complex conjugation of the coefficient vector causes no change in this inequality; either sign is covered by Drappeau's statement (or reflection and replacement of χ by $\bar{\chi}$). For $b_n(s) = \lambda \hat{\phi}(\lambda n) |n|^{-s}$, at $\Re s = \sigma_0$ or 0, the block norm is $\ll_B \lambda^2 H (1 + \lambda H)^{-B}$. Summing the geometric tails gives, with $Z = q^{1/2} M^{-1} \lambda^{-1-\varepsilon}$,

$$\mathcal{S}(K; b(s)) \ll L_*^2 \lambda (K^2 + Z). \quad (5.8)$$

At $\Re s = -1$, the coefficients of $\tilde{B}$ have modulus $\pi Y |n| |\hat{\phi}(\lambda n)|$; their bound is (5.8) multiplied by $(Y/\lambda)^2 \leq 1$, after increasing the decay exponent. This also proves the claims when $\lambda > 1$: then every nonzero mode lies in the rapidly decreasing tail.

For the varying vectors $b_n(\pm it_j)$, on each unit interval $[k, k+1]$ use the elementary inequality

$$|B_j(\pm it)|^2 \leq 2|B_j(\pm ik)|^2 + 2 \int_k^{k+1} |\partial_{\tau} B_j(\pm i\tau)|^2 d\tau.$$

Differentiating multiplies a coefficient by $\log|n|$, bounded on a block by $O(L_* + |\log(\lambda H)|)$; the extra factors are absorbed in the same block sums. Consequently the sum on $K < |t_j| \leq 2K$ for these vectors is $\ll K L_*^4 \lambda (K^2 + Z)$. Insert these bounds in (5.7) and the small- t estimate. Dyadic summation over $K \geq 1$ converges, since the integral term has weight K^{-4} and the residue term has arbitrary weight K^{-B} . Vertical integrals converge by the decay of $\mathcal{W}$. We obtain

$$\sum_j |\langle P_{\chi}, u_j \rangle|^2 \ll L_*^C \frac{\lambda}{Y} (1 + Z).$$

No truncation at logarithmic frequency or height was made.

The continuous spectrum and its zero mode. For nonzero modes the identical argument uses the continuous version of (5.2). In the unit-interval argument freeze the Eisenstein parameter in the Fourier coefficients and differentiate only the auxiliary variable in $|n|^{-i\tau}$, then evaluate on the diagonal $\tau = t$; no derivative of an Eisenstein coefficient is required. The constant term at infinity is $\delta_{\infty} y^{1/2+it} + \Phi_{\infty}(1/2 + it) y^{1/2-it}$. Its pairing with the box is

$$\lambda \hat{\phi}(0) Y^{-1/2} \{ \delta_{\infty} Y^{-it} \mathcal{W}(it) + \overline{\Phi_{\infty}(1/2 + it)} Y^{it} \mathcal{W}(-it) \}.$$

Unitarity of the scattering matrix on the unitary axis gives $\sum_c |\Phi_{c\infty}(1/2 + it)|^2 = 1$. Squaring, integrating and summing therefore costs $O(\lambda^2/Y)$, independently of the number of cusps. The cross term with nonzero modes is absorbed by $|a + b|^2 \leq 2|a|^2 + 2|b|^2$. Finally (5.5) averages exactly h_q character bounds; there is no loss from their number. Enlarging $\lambda^{-1-\varepsilon}$ to $(L_*/\lambda)^{1+\varepsilon}$ proves (5.4). $\square$

Without (SEL), the terms with $t_j = i\sigma_j$ remain. On a block $|n| \asymp H$, the Bessel factor has additional size $(HY)^{-\sigma_j}$. Thus (5.3) is used with $X = (HY)^{-1}$ when $HY \leq 1$, not with $X = Y^{-1}$. It does not recover Proposition 5.1 in the required cells; Section 9 records precisely the resulting unconditional limitation. Neither the Sobolev argument nor the regular-spectrum large sieve depends on (SEL).

6. PER- d COUNTS

Fix $c \geq 1$ and use $\mathcal{F}_d^I, \Gamma'_q$ from §3, with $n(Q) = cB - A = 4acd - f$. In $[A, B, C]$, A is a form coefficient, not the dyadic parameter; the parity and sieve invariance were proved there.

For a prime $\ell \nmid 2d$, put $\chi_d(\ell) = (-d/\ell)$ and define $g_{c,d}(\ell)$ as the proportion on the quadric $B^2 - 4AC = -4d$ satisfying $cB - A = 0$. Directly,

$$g_{c,d}(\ell) = \begin{cases} (\ell - 1)/(\ell^2 + \chi_d(\ell)\ell), & \ell \nmid c, \\ (1 + \chi_d(\ell))/(\ell + \chi_d(\ell)), & \ell \mid c. \end{cases} \quad (6.1)$$

Extend multiplicatively to squarefree q coprime to $2d$. The denominator counts the quadric: for $A \neq 0$ there are $\ell(\ell - 1)$ choices, while $A = 0$ gives $\ell(1 + \chi_d(\ell))$. If $\ell \nmid c$, the imposed equation forces $B \neq 0$, giving exactly $\ell - 1$ choices. If $\ell \mid c$, it forces $A = 0$, giving the second numerator. Primes dividing $2d$ never divide n : both f and n are odd, and f is a unit at every prime dividing d .

Lemma 6.1 (Factorisation of the orbital main term). *PROVED. Let $\#\Lambda_d(q)$ be the orbifold count of $\{Q \in \mathcal{F}_d^I : q \mid n(Q)\}$ modulo Γ'_q , and put $V(q) = \text{vol}(\Gamma'_q \backslash \mathbb{H})$. For squarefree $(q, 2d) = 1$,*

$$\frac{\#\Lambda_d(q)}{V(q)} = g_{c,d}(q) \frac{\#\Lambda_d(1)}{V(1)}.$$

Proof. There is nothing to prove for $q = 1$. Reduction of Γ'_1 onto $\text{SL}_2(\mathbb{Z}/q\mathbb{Z})$ is surjective: it contains $\Gamma(2d)$, and the assertion follows from CRT and surjectivity of reduction for $\text{SL}_2(\mathbb{Z})$. The kernel is Γ'_q . Thus the index of effective groups is $I_q = |\text{SL}_2(\mathbb{Z}/q\mathbb{Z})|/2$, since $-I$ belongs to Γ'_1 but not to Γ'_q . At a prime $\ell \mid q$, the stabiliser of a nondegenerate binary form in $\text{SL}_2(\mathbb{F}_\ell)$ is its special orthogonal torus, of order $\ell - \chi_d(\ell)$: it is either the split torus or the norm-one subgroup of $\mathbb{F}_{\ell^2}^\times$. Its orbit consequently has $\ell^2 + \chi_d(\ell)\ell$ elements, the size of the entire quadric. The fraction satisfying the sieve condition is therefore $g_{c,d}(\ell)$, independently of the original orbit. CRT gives the product of these fractions. For an orbit with stabiliser order e_Q , the weighted number of sieved Γ'_q -orbits above it is $e_Q^{-1} I_q g_{c,d}(q)$, by orbit-stabiliser (equivalently, sum over the cosets before dividing by the stabiliser). Summing over the original orbits, and using $V(q) = I_q V(1)$, proves the formula. $\square$

Choose a fixed physical box, so that in the coordinates $u = w/(2q)$ its weights are $\psi_q(u) = \psi_1(qu)$. Accordingly $\lambda_q = \lambda_1/q$, $Y_q = Y_1/q$, and $\int \psi_q d\mu = \int \psi_1 d\mu$. Define

$$\mathfrak{M}_d = \frac{\#\Lambda_d(1)}{V(1)} \int_{\mathbb{H}} \psi_q d\mu, \quad \kappa_c(q) = 2^{\omega((q,c))}.$$

The first quantity is independent of the sieve modulus. Here and below $\omega(m)$ counts the distinct prime factors of m .

Theorem 6.2 (Per- d count). *CONDITIONAL on (SEL). Let q be squarefree, $(q, 2d) = 1$, and let ψ_q be a box as in Proposition 5.1, with $\lambda_q \asymp A/(qF)$, $Y_q \asymp 1/(qF\sqrt{d})$, $A, F \geq 1$. Assume explicitly $Y_q \leq \lambda_q L_*^{-3}$ and polynomial parameter ranges in N , so $L_* \ll \mathcal{L}$. Then, for $0 < \varepsilon < 1/2$,*

$$\left| \sum_{\substack{Q \in \mathcal{F}_d^I \\ q|n(Q)}} \psi_q(u_Q) - g_{c,d}(q) \mathfrak{M}_d \right| \ll_{\varepsilon} \mathcal{L}^C q \kappa_c(q)^{1/2} (\#\Lambda_d(1))^{1/2} \left\{ A\sqrt{d} \left(1 + \frac{A}{qF} \right) + \frac{F^{1+\varepsilon}}{\sqrt{d}} \right\}^{1/2}. \quad (6.2)$$

In particular the factor $\kappa_c(q)^{1/2}$ is absent when $(q, c) = 1$, as in the sieve application. There is no lower bound on F/A .

Proof. Corollary 4.4 applies to the sieved subset, because separation is preserved by passage to a subset and by the dilation to u . Lemma 6.1 identifies its mean as $g_{c,d}(q) \mathfrak{M}_d$. In Proposition 5.1,

$$\frac{\lambda}{Y} \asymp A\sqrt{d}, \quad \frac{\lambda^2}{Y} \asymp A\sqrt{d} \frac{A}{qF}, \quad \frac{q^{1/2}}{MY} \lambda^{-\varepsilon} \ll \frac{F^{1+\varepsilon}}{\sqrt{d}} q^{-1/2+\varepsilon} A^{-\varepsilon} \leq \frac{F^{1+\varepsilon}}{\sqrt{d}}.$$

The remaining factor from Sobolev duality is $(\#\Lambda_d(q))^{1/2}$. For $\ell \nmid c$,

$$|\mathrm{SL}_2(\mathbb{F}_\ell)| g_{c,d}(\ell) = (\ell - \chi_d(\ell))(\ell - 1) \leq \ell^2;$$

for $\ell \mid c$ it is either zero or $2\ell(\ell - 1) \leq 2\ell^2$. The exact identity from Lemma 6.1 therefore gives $\#\Lambda_d(q) \leq \kappa_c(q) q^2 \#\Lambda_d(1)$, also for $q = 1$. Taking square roots proves (6.2). $\square$

The term $A/(qF)$ is the cost of wrapping around several periods; it carries no N^ε loss. One may instead use the e -cusp: $w \mapsto -1/(dw)$ sends the point to $(2a + i/\sqrt{d})/e$, and reflection gives the same boxes. Interchanging e, f replaces the sieve equation $A = cB$ by $C/d = cB$. The involution $(A, B, C) \mapsto (C/d, B, dA)$ is a bijection of the quadric modulo every prime not dividing $2d$; hence the densities and the estimate are unchanged.

Lemma 6.3 (Class-number bound). *PROVED. With $h(-4d)$ the number of primitive proper classes of discriminant $-4d$, and $r(d) = \prod_{p^k \parallel d} p^{\lfloor k/2 \rfloor}$,*

$$\#\Lambda_d(1) \leq 6h(-4d)r(d), \quad \sum_{d \leq D} \#\Lambda_d(1) \ll D^{3/2} \log^2(2D).$$

Proof. Every Type I form is primitive: a prime dividing f divides neither $2ad$ nor $4ad$, by $ef - 4a^2d = 1$. For a primitive class represented by Q_0 , its admissible level- d cosets are bounded by the number of $(t : s) \in \mathbb{P}^1(\mathbb{Z}/d\mathbb{Z})$ with $Q_0(t, s) \equiv 0 \pmod{d}$. Indeed, the second column of a matrix representing a coset modulo $\Gamma^0(d)$ gives this projective point; the condition that the transformed last coefficient be divisible by d is necessary. Passing from $\Gamma_0(d)$ to Γ'_1 costs index at most six (exactly six for odd d , four for even d); imposing Type I parity can only decrease the count. Stabiliser weights only improve this upper bound.

For $p^k \parallel d$, make an invertible change of variables so that $Q_0 = [A_0, B_0, C_0]$ has $p \nmid A_0$. This is possible by primitivity; at 2, B_0 is even and one of the outer coefficients is odd. Points with $p \mid s$ cannot be zeros. At an odd prime, zeros $(t : 1)$ satisfy $(2A_0t + B_0)^2 \equiv 0 \pmod{p^k}$, since the discriminant is $-4d$. There are exactly $p^{\lfloor k/2 \rfloor}$ possibilities. At 2, write $B_0 = 2B'_0$; the identity

$$A_0Q_0(t, 1) = (A_0t + B'_0)^2 + d$$

shows that the condition is $A_0t + B'_0 \equiv 0 \pmod{2^{\lceil k/2 \rceil}}$, again giving $2^{\lfloor k/2 \rfloor}$ possibilities. CRT proves the first assertion. The classical imaginary-quadratic class-number formula [IK] is $h(D_0) = w(D_0)\sqrt{|D_0|}L(1, \chi_{D_0})/(2\pi)$, where $w(D_0) \leq 6$ and χ_{D_0} is the Kronecker character, possibly imprimitive. Its nonprincipal character sums are bounded by $|D_0|$; splitting the Dirichlet series there and using partial summation gives $L(1, \chi_{D_0}) \ll \log(2|D_0|)$. Thus $h(-4d) \ll \sqrt{d}\log(2d)$, including nonfundamental discriminants. Finally,

$$\sum_{d \leq D} r(d) \leq \sum_{m \leq \sqrt{D}} m \#\{d \leq D : m^2 \mid d\} \leq D \sum_{m \leq \sqrt{D}} \frac{1}{m} \ll D \log(2D).$$

Multiplying by $\sqrt{D}\log(2D)$ proves the second assertion. $\square$

There is no uniform per- d relative-error assertion: the orbital main term contains a possibly small class number. Section 8 uses only absolute errors and $\sum_{d \lesssim D} \sqrt{\#\Lambda_d(1)} \ll D^{5/4} \log(2D)$, by Cauchy-Schwarz and Lemma 6.3. Relative to the box mass AD , the first summed error in (6.2) has size $\mathcal{L}^C q(D/A)^{1/2}(1 + A/(qF))^{1/2}$ when $(q, c) = 1$. This is an averaged absolute-error calculation, not a lower bound for an individual $\mathfrak{M}_d$.

7. THE PER- a WEIL COUNT

The complementary estimate is unconditional. It counts a modular hyperbola rather than a packet of forms. Write $e(t) = \exp(2\pi it)$ in this section, and put

$$S(h, k; m) = \sum_{x \bmod m}^* e((hx + k\bar{x})/m).$$

All smooth weights below have fixed derivative bounds. For upper bounds we take them nonnegative, with positive integral.

Proposition 7.1 (the count (K_a) ; PROVED). *Let $a, c \geq 1$, let q be squarefree with $(q, 2a) = 1$, and set $m = 4a^2$, $r = mq$. For $D, E \geq 1$ put $F = mD/E$, and let $W_1, W_2 \in C_c^\infty((1, 2))$. Define*

$$S_a(q) = \sum_{\substack{e, f \geq 1 \\ ef \equiv 1 \pmod{m} \\ q \mid 4acd - f}} W_1(e/E)W_2(d/D), \quad d = (ef - 1)/m.$$

Then $S_a(q) = g'_{c,a}(q)X_a + R_a(q)$, where

$$X_a = \frac{\varphi(m)}{m^2} \iint_{\mathbb{R}^2} W_1(x/E)W_2((xy - 1)/(mD)) dx dy \asymp D \frac{\varphi(m)}{m}, \quad (7.1)$$

$$|R_a(q)| \ll 2^{\omega(q)} \tau(m)^2 \left(qa + \frac{D}{E} + \frac{D}{F} \right), \quad (7.2)$$

and $g'_{c,a}$ is multiplicative on squarefree numbers, with

$$g'_{c,a}(\ell) = \begin{cases} (\ell - 1)/\ell^2, & \ell \nmid c, \\ 1/\ell, & \ell \mid c. \end{cases}$$

The constants depend only on the weights.

Proof. The identity

$$an = f(ce - a) - c$$

shows that the residue conditions on (e, f) modulo r are the CRT product of

$$\mathcal{S}_m = \{(e, f) : ef = 1 \pmod{m}\}, \quad \mathcal{S}_q = \{(e, f) : f(ce - a) = c \pmod{q}\}.$$

Put $W(x, y) = W_1(x/E)W_2((xy - 1)/(mD))$. On its support $x \asymp E$, $y \asymp F$, and $\partial_x^i \partial_y^j W \ll_{i,j} E^{-i} F^{-j}$. Two-dimensional Poisson summation gives

$$S_a(q) = r^{-2} \sum_{h,k \in \mathbb{Z}} \widehat{W}(h/r, k/r) \mathcal{K}(h, k), \quad (7.3)$$

$$|\widehat{W}(h/r, k/r)| \ll_B EF(1 + E|h|/r)^{-B}(1 + F|k|/r)^{-B}.$$

The sign in the finite Fourier transform is immaterial; with the positive sign, CRT gives

$$\mathcal{K}(h, k) = S(\bar{q}h, \bar{q}k; m)T_q(\bar{m}h, \bar{m}k), \quad T_q(u, v) = \sum_{(e,f) \in \mathcal{S}_q} e((ue + vf)/q).$$

The prime factors of T_q have CRT-rescaled arguments. For a prime $\ell \mid q$, if $\ell \nmid c$, the substitution $w = ce - a$ gives

$$T_\ell(u, v) = e(u\bar{c}a/\ell)S(u\bar{c}, v\bar{c}; \ell).$$

If $\ell \mid c$, then $f = 0$ and $T_\ell(u, v) = \ell \mathbf{1}_{\ell|u}$. In particular $|T_q(u, v)| \leq 2^{\omega(q)}q$ (the trivial cardinality bound suffices). At $(h, k) = (0, 0)$ we have $|\mathcal{S}_m| = \varphi(m)$ and $|\mathcal{S}_\ell| = \ell - 1$ or ℓ , respectively. This gives exactly (7.1) times $g'_{c,a}(q)$.

We retain the axial frequencies: they are important if a divisor is small. For $B > 2$ and every integer $v \geq 1$,

$$\sum_{\substack{k \neq 0 \\ v|k}} (1 + F|k|/r)^{-B} \ll_B \frac{r}{vF}. \quad (7.4)$$

Indeed, if $vF \leq r$, compare with an integral; otherwise the left side is $O_B((r/(vF))^B)$. Since $|S(0, \bar{q}k; m)| = |c_m(k)| \leq (k, m)$, expanding $(k, m) \leq \sum_{v|(k,m)} v$ in (7.4) gives

$$\sum_{k \neq 0} (k, m)(1 + F|k|/r)^{-B} \ll \tau(m)r/F.$$

Thus the frequencies $h = 0 \neq k$ contribute at most $2^{\omega(q)}\tau(m)E/m = 2^{\omega(q)}\tau(m)D/F$; $k = 0 \neq h$ contributes at most $2^{\omega(q)}\tau(m)D/E$.

For $hk \neq 0$ use the classical Weil–prime-power bound [Weil], in the convenient form

$$|S(u, v; m)| \leq 3\tau(m)m^{1/2}(u, v, m)^{1/2}.$$

The constant allows the prime 2. Expanding the gcd as before yields

$$\begin{aligned} & \sum_{h,k \neq 0} (h,k,m)^{1/2} (1 + E|h|/r)^{-B} (1 + F|k|/r)^{-B} \\ & \leq \sum_{v|m} v^{1/2} O_B \left(\frac{r}{vE} \frac{r}{vF} \right) \ll_B \frac{r^2}{EF}, \end{aligned}$$

since $\sum_{v \geq 1} v^{-3/2} < \infty$. This remains valid when $vE > r$ or $vF > r$. Inserting it in (7.3) bounds the nonaxial contribution by $O(2^{\omega(q)} q \tau(m) m^{1/2})$. As $m^{1/2} = 2a$, the claimed error follows, even with one power of $\tau(m)$ here. $\square$

For later use, if $\min(E, F) \geq N^{c_0}$, summation over $a \asymp A$ and squarefree $q \leq Q$ gives

$$\sum_{a \asymp A} \sum_{\substack{q \leq Q \\ (q, 2a)=1}} \mu^2(q) 3^{\omega(q)} |R_a(q)| \ll \mathcal{L}^C (Q^2 A^2 + QADN^{-c_0}). \quad (7.5)$$

Here we used the elementary divisor-moment bound $\sum_{a \asymp A} \tau(4a^2)^2 \ll A(\log(2A))^C$. Measured against the scale AD , the error is $\mathcal{L}^C (Q^2 A/D + QN^{-c_0})$. This gives a useful sieve level on the side $D > A$. The axial errors explain why a uniform density assertion with no lower bound on E, F would be false. No small-divisor substitute for this proposition is used below.

8. ASSEMBLY

We begin with the elementary accounting fact that permits a sieve level to approach 1 at the boundary $A = D$.

Lemma 8.1 (linearly degenerating savings; PROVED). *Let $L \geq 2$, and let $m_{jk} \geq 0$, $1 \leq j, k \leq \lfloor L \rfloor$, with $\sum_k m_{jk} \leq M$ for each j . Then*

$$\sum_{j,k} m_{jk} \min(1/j, 1/k) \leq M(1 + \log L).$$

If also $m_{jk} \leq M/L$, the same sum is at most $2M$.

Proof. The first assertion follows by using $1/j$ and summing the harmonic series. For the second, group pairs by $h = \max(j, k)$:

$$\sum_{j,k \leq L} \min(1/j, 1/k) = \sum_{h \leq L} \frac{2h-1}{h} \leq 2L.$$

Multiply by M/L . Fixed constant multiples of either saving only change the resulting absolute constant. $\square$

Below, j indexes a c -block and k its distance from $A = D$. Layer masses are $O(N\mathcal{L})$, so take $M \asymp N\mathcal{L}^2$ and $L \asymp \mathcal{L}$. The block $j = 0$ is treated separately, never with $1/j$.

Theorem 8.2 (CONDITIONAL on (SEL)). *Assume (SEL) as stated in Section 1, uniformly for $\Gamma_0(4dq^2)$ and the even nebentypus characters modulo $2q$. Then*

$$\sum_{p \leq N} f_I(p) \ll N \log^2 N.$$

Proof. We work with $N/2 < p \leq N$ and write $\mathcal{L} = \log N$. The reduction of Section 2, from [ET, Proposition 2.2 and Lemma 2.8], gives $f_I(p) \leq 2 \sum_c w_c(p)$. Fix small constants $0 < \eta \leq \eta_1$; all subsequent constants may depend on them.

1. *Removing the short-progression ranges.* Proposition 2.8 bounds the part $c > N^\eta$ by $O_\eta(N\mathcal{L}^2)$. For $c \leq N^\eta$, Lemma 2.6 disposes, all c at once, of tuples with

$$\min(4ab, 4acf, 4cdf) \leq N^{1-\eta_1}.$$

Indeed $bf = na + c$ and $f \leq 2n$ imply $b \geq a/2$, so $4ad$ and $4bd$ are already $\gg N^{1-\eta}$; no other short modulus is needed. The remaining tuples lie in $\mathcal{R}_{\text{bad}}(\eta_1)$ up to $O(1/\mathcal{L})$. Use nonnegative smooth cells of bounded overlap supported in $\mathcal{R}_{\text{bad}}(2\eta_1)$. Write

$$\begin{aligned} c &= N^\gamma, & a &\asymp A = N^\alpha, & d &\asymp D, & AD &\asymp N/c, \\ e &\asymp E = N^{\beta-\gamma}, & f &\asymp F = N^{1+\alpha-\beta}. \end{aligned} \quad (8.1)$$

Cells have side $O(1/\mathcal{L})$ in the exponent variables. The spectral weights may instead be chosen in $(f', a/f')$, $f' \in \{e, f\}$; these coordinates give the same bounded-overlap covering. Positive majorants allow us to drop the original size restrictions inside a cell. Proposition 2.3 gives, throughout the enlarged bad region,

$$e, f \gg N^{1/2-3\eta_1}. \quad (8.2)$$

For example $\beta \geq (1 - 2\eta_1)/2 + O(1/\mathcal{L})$, while $1 + \alpha - \beta \geq \max(1 - \alpha - \gamma - 2\eta_1, \alpha - 2\eta_1) + O(1/\mathcal{L})$.

2. *Bands and layers.* Let $c \asymp 2^j$, $0 \leq j \ll \eta\mathcal{L}$, and put

$$\delta = |2\alpha - 1 + \gamma|, \quad k = \lfloor \delta\mathcal{L} \rfloor.$$

A layer means fixed (j, k) and all its β -cells; there are only $O(1)$ choices of the a -block for each c -block and k . Fixed shifts in j, k and fixed changes of logarithm base are absorbed in constants. Remove bands of width $C_0/\mathcal{L}$ about $\beta = \alpha + \gamma$ and $\beta = 1$, and also the cells with $D \leq 64$. In each such cell one of e, f is $O_{C_0}(A)$ (for the last band use $ef = 4a^2d + 1$). Lemma 2.7(b), with a fixed enlarged constant in $F \leq 4A$, bounds its mass by $O(2^j AD) = O(N)$. There are $O(\mathcal{L})$ band cells per c -block, so all bands cost $O(N\mathcal{L}^2)$ without a sieve. Layers $k \leq C_1 \log \mathcal{L}$ are postponed.

For reference, Brun–Titchmarsh on $4ad$ bounds a whole layer by

$$O(N\mathcal{L}/j), \quad j \geq 1, \quad (8.3)$$

whenever $k \leq \mathcal{L}/3$. For fixed (a, d, f) the c -block maps into an interval of length $4ad2^j$ in one residue class modulo $4ad$. Brun–Titchmarsh [ET, (A.10)] gives $8ad2^j/(\varphi(4ad)j \log 2)$ for a reduced class. In a nonreduced class there is at most one prime, which is absorbed by the same upper bound. Now $k \leq \mathcal{L}/3$ implies $A, D \geq N^{1/4}$ for our fixed small η , and Lemma 2.7(a2), summed over the divisors f , proves (8.3). For $j = 0$ use its trivial analogue $O(N\mathcal{L})$.

3. *The cell estimates away from the bands.* All errors in the next paragraphs are summed absolute errors normalized by AD (by N after the c -sum), using Cauchy–Schwarz and Lemma 6.3 for Theorem 6.2. The bad region gives $A \geq N^{1/3-2\eta_1}$, hence $A \geq \mathcal{L}^2$ for Lemma 2.7(a1). All spectral boxes satisfy $Y/\lambda \asymp (A\sqrt{d})^{-1} \ll N^{-1/3+2\eta_1}$ and $L_* \ll \mathcal{L}$, so Proposition 5.1’s exact height condition holds for large N . We write F' for the scale of the selected divisor.

(i) $D \geq A$. Use the fixed- a sequences of Proposition 7.1. By (7.5) and (8.2) their normalized error, summed with sieve weights through $q \leq Q$, is

$$\ll \mathcal{L}^C (Q^2 N^{-\delta} + Q N^{-c_0}), \quad c_0 = \frac{1}{2} - 4\eta_1 > 0.$$

There is no restriction on β . The bad region gives $\alpha \geq 1/3 - 2\eta_1$, hence $\delta \leq 1/3 + 4\eta_1$ here. Thus $Q \leq N^{\delta/4}$ makes the second term an absolute power saving, for sufficiently small η_1 .

(ii) $D < A$ and $\beta < \alpha + \gamma$. Here $e \ll A$. If $k \geq 2j$ (allowing fixed shifts of the thresholds), use the e -cusp variant of Theorem 6.2, whose density is unchanged by the involution proved in §6. Thus $F' = E$ and $\lambda \asymp A/(qE)$, including $\lambda > 1$. Its summed error is

$$\ll \mathcal{L}^C Q^2 \left[N^{-\delta/2} (1 + A/E)^{1/2} + \frac{E^{1/2+\varepsilon}}{A} \right]. \quad (8.4)$$

Since $b \geq a/2$, we have $\beta \geq \alpha - O(1/\mathcal{L})$ and $(A/E)^{1/2} \ll N^{\gamma/2} \ll N^{\delta/4}$ in this range. The other term is $\ll A^{-1/2+2\varepsilon}$ and is $O(N^{-\delta/4})$ on the bad region, for a fixed sufficiently small ε . This gives $\mathcal{L}^C Q^2 N^{-\delta/4}$. Cells with $a/2 \leq b < a$ are included: they satisfy $e \leq 3a/c$. If $k < 2j$, use (8.3) instead; its hypotheses hold because $j \ll \eta \mathcal{L}$.

(iii) $D < A$ and $\alpha + \gamma < \beta < 1$. Off the bands, both $e, f \geq 8A$. Choose the smaller divisor, with scale $F' = \min(E, F) \ll A\sqrt{D}$, as cusp parameter. Then $\lambda \ll 1/q$, and Theorem 6.2 gives

$$\ll \mathcal{L}^C Q^2 \left[N^{-\delta/2} + N^\varepsilon D^{1/4} A^{-1/2} \right].$$

The second exponent is $(1 - \gamma - 3\alpha)/4 + \varepsilon$; its difference from $-\delta/4$ is $-\alpha/4 + \varepsilon$. It is therefore bounded by $N^{-\delta/4}$ with an absolute margin. We use the weaker common error $\mathcal{L}^C Q^2 N^{-\delta/4}$.

(iv) $D < A$ and $1 < \beta \leq 1 + 2\eta_1$. This strip cannot be discarded: here $f < A$, even though e is large. Put $k' = \lfloor (\beta - 1)\mathcal{L} \rfloor$, which exceeds a fixed constant off the band. If $k' \leq k/2$, use the f -cusp, not the e -cusp. With $F' = F$ and $\lambda \asymp A/(qF)$, Theorem 6.2 gives

$$\ll \mathcal{L}^C Q^2 \left[N^{-\delta/2} (1 + A/F)^{1/2} + F^{1/2+\varepsilon}/A \right] \ll \mathcal{L}^C Q^2 N^{-\delta/4},$$

because $(A/F)^{1/2} \ll N^{(\beta-1)/2} \ll N^{\delta/4}$. The cusp term has an absolute margin. Crucially, the first term carries no N^ε loss at the degenerating boundary. If $k' > k/2$, use the cdf fibres of Lemma 2.6: $4cdf \asymp N^{2-\beta}$ and $\log(N/(4cdf)) \gg k'$. Brun-Titchmarsh costs $O(N/k')$ per cell. Indeed its reciprocal-totient sum is bounded by

$$\sum_{c \asymp 2^j} \frac{1}{\varphi(c)} \sum_{d \asymp D} \frac{1}{\varphi(d)} \sum_{f \asymp F} \frac{\rho_d(f)}{\varphi(f)} \ll 1$$

by Lemma 2.5(c). Comparing with this weighted sum bounds the nonreduced-class errors by $\sum_{c,d,f} \rho_d(f) \ll 2^j DF \asymp N \exp(-k') \ll N/k'$. On the sieve side the weighted cell mass is $O(N)$ by Lemma 2.7(b), since $F \leq A$. Thus in either treatment the cell costs $O(N/\max(k, k'))$ once the sieve saving below is established.

These cases cover every remaining cell: on $D \geq A$ use (i), and on $D < A$ the three intervals separated by $\alpha + \gamma$ and 1 give (ii)–(iv). Their boundaries and $A \asymp D$ were already set aside.

4. *Obtaining the prime saving from the cell errors.* For a sequence σ with fixed $s \in \{a, d\}$ and fixed c , choose

$$z = N^{\kappa\delta/8}, \quad Q = z^2, \quad \kappa = 1 \text{ in (i)}, \quad \kappa = 1/4 \text{ in (ii)–(iv)}.$$

Sieve only by primes $\ell > \ell_0 \geq 3$ with $\ell \nmid 2cs$. Thus $(q, 2ca) = 1$ in Proposition 7.1, and $(q, 2cd) = 1$ in Theorem 6.2; in particular $\kappa_c(q) = 1$. Both densities satisfy

$$\frac{1}{\ell} - \frac{2}{\ell^2} \leq g_\sigma(\ell) \leq \frac{1}{\ell} < 1.$$

Selberg's upper-bound sieve [HR, Theorem 4.1 and Lemma 4.1] for nonnegative weighted sequences therefore gives

$$\begin{aligned} \#\{n \in \sigma : n \text{ prime}, n > N/2\} &\leq \frac{X_\sigma}{G_\sigma(z)} + \sum_{\substack{q \leq Q \\ q|P_\sigma(z)}} 3^{\omega(q)} |r_\sigma(q)|, \\ G_\sigma(z) &\gg \frac{\varphi(2cs)}{2cs} \log z. \end{aligned} \tag{8.5}$$

Here counts are weighted if the cell is smooth; $z < N/2$. The lower bound on the local density, not merely its upper bound, is used in the estimate for G_σ .

The model mass X_σ need not be compared from below to the actual mass. At $q = 1$, Theorem 6.2 or Proposition 7.1 gives

$$X_\sigma \leq \#\sigma + |r_\sigma(1)|.$$

Also $2cs/\varphi(2cs) \leq 2g(c)g(s)$. Lemma 2.7(a1) and $\sum_{c \lesssim 2^j} g(c) \ll 2^j$ bound the weighted mass of a layer in (i) or (iii) by $O(N\mathcal{L})$. Lemma 2.7(b) bounds each cell in (ii) or (iv) by $O(N)$; in particular their layer masses are $O(N\mathcal{L})$. The $q = 1$ errors introduced in this comparison have only the extra factor $\max g(c)g(s) \ll (\log \mathcal{L})^2$. Since $Q^2 = N^{\kappa\delta/2}$, the summed errors are $\ll \mathcal{L}^C N^{-\kappa\delta/2}$ times the relevant mass scale; the additional term in (i) has an absolute power saving. All these errors are $O(1/k)$ times that scale once $k \geq C_1 \log \mathcal{L}$, with C_1 fixed sufficiently large. The main term in (8.5) has the same saving, since $\log z \asymp k$. This proves all the sieve savings asserted in step 3.

5. *Summing all cells.* For $1 \leq j \ll \eta\mathcal{L}$, the layers $k \leq C_1 \log \mathcal{L}$ cost, by (8.3),

$$\ll N\mathcal{L} \log \mathcal{L} \sum_{j \ll \eta\mathcal{L}} \frac{1}{j} \ll N\mathcal{L}(\log \mathcal{L})^2 \ll N\mathcal{L}^2.$$

For $j = 0$ these layers cost $O(N\mathcal{L} \log \mathcal{L})$ trivially. In (i), (iii), and (ii) with $k \geq 2j$, a layer costs

$$\ll N\mathcal{L} \min(1/j, C/k) \quad (k \leq \mathcal{L}/3).$$

One takes the better bound for the whole portion of the layer in question, not an unweighted average of the two methods. Lemma 8.1 sums these costs to $O(N\mathcal{L}^2)$. For $k > \mathcal{L}/3$ the sieve alone suffices: $\sum_j \sum_{k > \mathcal{L}/3} N\mathcal{L}/k \ll N\mathcal{L}^2$. Case (ii) with $k < 2j$ costs $\sum_j \sum_{k < 2j} O(N\mathcal{L}/j) = O(N\mathcal{L}^2)$. In (iv) the per-cell bound gives, for each c -block,

$$\sum_{k, k' \ll \mathcal{L}} \frac{CN}{\max(k, k')} \ll N \sum_{h \ll \mathcal{L}} \frac{2h-1}{h} \ll N\mathcal{L}.$$

Summing over j gives $O(N\mathcal{L}^2)$ again. For $j = 0$ outside the innermost layers the $1/k$ bound costs at most $O(N\mathcal{L} \log \mathcal{L})$, and the preceding two-dimensional sum still

applies to (iv). Together with the bands and step 1, this proves the bound on $(N/2, N]$. Summing dyadically in N proves the theorem. $\square$

9. WHAT IS UNCONDITIONAL

The separation argument, Sobolev duality, the local-density computation, the per- a count, and the arithmetic mass estimates are unconditional. The use of (SEL) is confined to the exceptional spectrum in Proposition 5.1. It is nevertheless essential to the present proof of the prime-counting theorem.

9.1. The exceptional-spectrum obstruction. An eigenvalue $\lambda_j = 1/4 - \sigma_j^2$, $0 < \sigma_j < 1/2$, introduces in the unfolded coefficient the factor $(nY)^{-\sigma_j}$, since $K_{\sigma_j}(2\pi ny)$ grows like a constant times $(ny)^{-\sigma_j}$ when ny is small. The unconditional Kim–Sarnak bound [KS, Appendix 2] is $\sigma_j \leq 7/64$, including the nebentypus forms at issue here. A direct use of this bound permits the loss

$$Y^{-7/64} \asymp (qF'\sqrt{d})^{7/64} \quad (9.1)$$

in the spectral error. On the $D < A$ side this must be compared with $(D/A)^{1/2} = N^{-\delta/2}$, $\delta = 2\alpha - 1 + \gamma > 0$. In the worst cells $qF'\sqrt{d}$ has size about N , so the exponent comparison leaves a strip

$$0 < 2\alpha - 1 + \gamma \lesssim 7/32.$$

Even the smallest cusp parameters at the boundary, with $F' \gg N^{1/2-3\eta_1}$ and D near $N^{1/2}$, give a loss of size $N^{21/256-O(\eta_1)}$, corresponding to the comparison $\delta \approx 21/128 - O(\eta_1)$. These numbers describe the obstruction to the estimates, not a sharp geometric boundary or a lower bound for an actual spectral contribution. In particular a fixed positive width, rather than a single line, remains untreated.

Assessment 9.1 (partial exceptional-spectrum estimates). Deshouillers–Iwaniec’s exceptional large sieve [DI, Theorem 5] improves on a pointwise insertion of (9.1), but does not supply the needed uniform saving at $A \asymp D$. On a Fourier block $n \asymp N_0$ its parameter must be

$$X = \frac{1}{N_0 Y},$$

not $1/Y$, because the squared exceptional weight is $(nY)^{-2\sigma_j}$. In its usual notation the bound has the shape

$$\sum_{\text{exc}} X^{2\sigma_j} \left| \sum_{n \asymp N_0} a_n \rho_j(n) \right|^2 \ll (1 + \sqrt{\mu N_0 X})(1 + \sqrt{\mu N_0^{1+\varepsilon}}) \sum_n |a_n|^2,$$

with $\mu \asymp 1/M$, $M = 4dq^2$. The first factor is of size $1 + (F'/(q\sqrt{d}))^{1/2}$, which is not uniformly harmless near the boundary. The level average in [DI, Theorem 6] offers further partial control. The source calculation reports an extra factor $(F')^{3/4} D^{1/8}/A$ there; its detailed bookkeeping was not re-checked in the internal review, and we do not use it as a proved obstruction theorem. The same caveat applies to the source’s exploratory exceptional-density calculations. None enters the conditional proof.

A positive-width region bounded only by the original $1/j$ saving still contributes at the level of the available upper bounds $N\mathcal{L}^2 \sum_{j \ll \mathcal{L}} 1/j$. Unlike the linearly degenerating saving in Lemma 8.1, this does not remove $\log \log N$. Thus *this argument gives no unconditional improvement* of [ET, Theorem 1.1]; this is not a

claim that an unconditional improvement by other methods is impossible. Sufficient replacements for (SEL) would include a large sieve for the exceptional spectrum with precisely these Fourier weights and $X \asymp 1/(N_0 Y)$, with errors that still allow the layer summation, or cancellation beyond termwise Weil in $\sum_{a \asymp A} S(h, k; 4a^2)$.

9.2. Lifts versus equidistribution of a finite packet. The lift-counting target (H^{**}) of [MN3, §3.5] is stronger than what the proof above uses. For the packets of natural class-number size $d^{1/2+o(1)}$ on a surface of volume $d^{1+o(1)}$, a box of hyperbolic area V has expected packet count $Vd^{-1/2+o(1)}$. Consequently a uniform relative equidistribution statement in *all* boxes cannot extend below area $d^{1/2-o(1)}$; integrality and boxes containing a packet point already prevent it. Here “scale” is area, not hyperbolic radius. This elementary granularity obstruction does *not* refute (H^{**}) as formulated for lifts, averaged in d , with congruence conditions. Nor does it obstruct the second-moment method here: the lift-counting Poincaré function has box area $\asymp A\sqrt{d}$, and Proposition 4.3 uses only its variance against a separated packet.

For comparison, Liu–Masri–Young [LMY, Theorem 1.4, and the more general Theorem 7.1] prove a joint level–discriminant equidistribution result in the range $q \leq |D|^{1/20-\varepsilon}$. Their q is a prime level, $-D$ is an odd fundamental discriminant, and q splits in $\mathbb{Q}(\sqrt{-D})$. Their count is in translates of a level-one fundamental domain; it is not the present thin-box count with sieve twists. Here level and discriminant are tied, already d and $-4d$ before inserting q , and the coprimality/splitting hypotheses do not match. Thus that theorem does not supply (H^{**}) or the needed per- d estimate. We make no exhaustive literature claim that no other joint-aspect theorem exists.

10. OPEN PROBLEMS

Problem 1 (unconditional removal). Prove $\sum_{p \leq N} f_I(p) \ll N \log^2 N$ without (SEL), or obtain an unconditional improvement of its present $\log \log N$ loss. A proof must address a positive-width strip adjacent to $A = D$, not merely the line $A = D$ itself.

Problem 2 (weighted exceptional density). Prove a density or large-sieve theorem for exceptional eigenvalues on $\Gamma_0(4dq^2)$, with even nebentypus modulo $2q$, incorporating the coefficients of the box Poincaré series and the weights $(N_0 Y)^{-2\sigma_j}$. The relevant target is an averaged error that retains a saving degenerating at most linearly in $|2\alpha - 1 + \gamma|$, after summation in d and the sieve modulus; a count of exceptional eigenvalues alone need not suffice.

Problem 3 (square-modulus Kloosterman sums). Obtain useful cancellation in $\sum_{a \asymp A} S(h, k; a^2)$, or its $4a^2$ variant, uniformly in the Fourier ranges produced by (7.3). For the application the averaging must handle the accompanying smooth weights, gcd factors, and the sieve congruence. Termwise square-root cancellation is exactly the estimate that stops at $D \asymp A$.

Problem 4 (joint level–discriminant counting). Establish a version of the lift-counting target (H^{**}) at level comparable to the discriminant, with the parity restriction and congruence twists retained. A second-moment theorem for the appropriate Poincaré functions could suffice and would avoid a false assertion of uniform fine-scale equidistribution of a finite packet.

Problem 5 (general numerators). Extend the conditional Type I estimate uniformly from $4/n$ to m/n . Primes dividing m must be omitted from the sieve, and the

mass gain $\varphi(m)/m$ must be tracked against the sieve-product inflation $m/\varphi(m)$ and all remainder terms [MN3, remarks after Theorem 3.8]. The earlier internal transition gap $(m \log m/\varphi(m))^{1/3}$ is historical: the refined lower-side bound in [MN3, §2, Theorem L'] reduces it to $(\log \log N)^{1/3}$. An m -uniform removal of the Type I logarithmic loss would close that remaining gap at the scale $\log N \asymp m^{1/3}$. Such an extension is not proved here.

Remark 10.1 (numerical status: EVIDENCE). The script `scripts/ttl_perd.py` supplies only a weak sanity check for per- d counts. Its weights are normalized to mass about 20, and its local-density comparison is not the orbital main term of Theorem 6.2. Small observed errors are therefore neither evidence for a uniform spectral error bound at the required scales nor a test of (SEL). No numerical observation is used in any proof.

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